The Graph–State Correspondence: A First-Principles Bridge From Compute-Graph Ollivier–Ricci Curvature to Quantum Dissipation

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Abstract

Phase C of the OpenMythosQuantum engine couples the Ollivier–Ricci curvature κ_e of a compute graph $G = (V, E)$ to the dissipation rates γ_e of a Lindblad master equation via the prescription $\gamma_e = \alpha \max(0, -\kappa_e)$. This coupling, while empirically successful, was introduced without physical justification: a classical information-geometric quantity on a discrete graph has no a priori reason to set the decoherence rate of a quantum state on a Hilbert space. We resolve this semantic gap. We define a class of hopping-type Lindbladians on $\mathcal{H}_G = \bigotimes_{v \in V} \mathbb{C}^2$ for which the Lindblad semigroup preserves the diagonal sector $\mathcal{D} \subset B(\mathcal{H}_G)$ and restricts there to a reversible continuous-time Markov chain whose generator is the weighted graph Laplacian of G with edge weights γ_e (the *Shadow Lemma*). Combining the Erbar–Maas entropic Ricci bound with the quantum H-theorem, we prove that for the Lindblad semigroup to have a uniform modified logarithmic Sobolev constant on the diagonal sector, the rates must satisfy $\gamma_e \geq \gamma_0 \max(0, -\kappa_e)$ for a universal γ_0 set by the graph's log-Sobolev constant. The Phase C formula is thus recovered as the *minimal* thermodynamically consistent coupling, with α identified as γ_0. We verify the correspondence numerically on the bowtie graph from the Phase C test suite, where we observe a sharp phase transition in the shadow-chain spectral gap at the analytically predicted threshold $\alpha_c = 1/|\kappa_{\text{bridge}}| = 3$. A universality study across seven graph families (single-bridge dumbbells K_n–K_n for $n = 3, 4, 5$; a path-graph control; a ladder-bridge topology; a series path-bridge; and an asymmetric K_3–K_5 dumbbell) shows that the prediction $\alpha_c = 1/|\kappa_{\min}|$ holds within a factor of 2 across all cases, with ratio $0.94 \leq \alpha_c^{\text{meas}}/\alpha_c^{\text{pred}} \leq 1.65$ and systematic drift governed by cluster size rather than topology class. Three falsifiable predictions follow: (i) mixing-time scaling $\tau \sim (\alpha - \alpha_c)^{-1}$ near threshold; (ii) a spectral-gap collapse on edges with $\gamma_e < |\kappa_e|$; and (iii) an entropy-production floor at the value set by the classical Ricci bound.

1 Introduction

In Phase C of the OpenMythosQuantum construction, the dissipation structure of an open quantum system is adaptively shaped by the local geometry of a classical compute graph. The coupling prescription

$$\gamma_e \;=\; \gamma_{\text{base}} \;+\; \alpha \max\bigl(0,\, -\kappa_e\bigr)^p, \qquad e \in E(G), \tag{1}$$

relates the Ollivier–Ricci curvature κ_e of each graph edge to the Lindblad jump-operator rate γ_e via a nonnegative, piecewise-linear response. Empirically, this mapping achieves everything we asked of it: the resulting semigroup is CPTP by construction, negatively-curved edges attract larger γ_e values, and the global dynamics remains physically consistent.

But the construction suffers from a semantic defect that no amount of numerical validation repairs. The Ollivier–Ricci curvature [1, 2] is an information-geometric quantity defined on a discrete metric space; it has no a priori connection to the decoherence rate of a quantum density matrix on a Hilbert space. The two objects live in different mathematical categories. Writing $\gamma_e \propto -\kappa_e$ is, strictly speaking, a heuristic with dimensional and structural content arbitrarily imposed.

A reviewer would correctly press on several points:

- What *is* the graph G, operationally? Which degrees of freedom of the quantum system do its nodes label?
- Why should *curvature* specifically drive dissipation? Node degree, betweenness, or the Laplacian spectrum are equally geometric.
- Why the functional form $\max(0, -\kappa_e)$ and not, say, $\exp(-\kappa_e)$ or $(1-\kappa_e)^2$?
- Why is the constant α free? A thermodynamically meaningful coupling should have a distinguished value, not an adjustable knob.

This paper provides a principled answer to all four questions. We construct a precise correspondence between the compute graph and the Lindblad dynamics (Section 2), prove a theorem showing that the Phase C mapping is the minimal thermodynamically consistent coupling (Section 3), and verify the predicted critical coupling numerically on the Phase C bowtie test case (Section 4). Three falsifiable predictions are extracted (Section 5); connections to tensor-network geometry, Lieb–Robinson bounds, and noncommutative Ricci curvature are discussed (Section 6).

The central mathematical object, which we call the *Shadow Lemma* (Proposition 1), identifies a distinguished subspace of the quantum state space on which the Lindblad semigroup reduces exactly to a classical Markov chain on G. This reduction is what licenses the curvature-dissipation identification: on the shadow sector, quantum entropy production becomes classical entropy production, and the Erbar–Maas Ricci bound applies verbatim.

2 The Graph–State Correspondence

2.1 Setup

Let $G = (V, E)$ be a finite, connected, undirected graph with $|V| = n$. To each vertex $v \in V$ we associate a local Hilbert space $\mathcal{H}_v \simeq \mathbb{C}^2$ with computational basis $\{|0\rangle_v, |1\rangle_v\}$. The total Hilbert space is

$$\mathcal{H}_G \;=\; \bigotimes_{v\in V} \mathcal{H}_v \;\simeq\; \mathbb{C}^{2^n}. \tag{2}$$

The basis states $\{|s\rangle : s \in \{0,1\}^V\}$ are called *computational configurations*. The *diagonal sector* is the commutative subalgebra

$$\mathcal{D} \;=\; \big\{\, \rho \in B(\mathcal{H}_G) \;:\; \langle s|\rho|s'\rangle = 0 \text{ for } s \neq s' \,\big\}. \tag{3}$$

Density operators $\rho \in \mathcal{D}$ are in one-to-one correspondence with probability distributions $\mu_\rho(s) = \langle s|\rho|s\rangle$ on $\{0,1\}^V$.

Definition 1 (Hopping jump operators)**.** For each edge $e = (u, v) \in E$, the *hopping jump operator* is

$$L_e \;=\; \sigma_u^- \otimes \sigma_v^+ \;+\; \sigma_u^+ \otimes \sigma_v^-, \tag{4}$$

acting as the identity on all other factors of $\mathcal{H}_G$. Here $\sigma^- = |0\rangle\langle 1|$ and $\sigma^+ = |1\rangle\langle 0|$.

The operator L_e swaps the excitation status of the two endpoints of edge e whenever they differ. It is Hermitian, $L_e^\dagger = L_e$, and $L_e^2 = P_e$ where P_e is the projector onto configurations where endpoints u, v have different values.

Definition 2 (Graph Lindbladian)**.** Given rates $\{\gamma_e \geq 0 : e \in E\}$, the *graph Lindbladian* on $\mathcal{H}_G$ with free Hamiltonian H is

$$\mathcal{L}[\rho] \;=\; -i[H,\rho] \;+\; \sum_{e\in E} \gamma_e \Big(L_e \rho L_e^\dagger - \tfrac{1}{2}\{L_e^\dagger L_e, \rho\}\Big)\,. \tag{5}$$

2.2 The Shadow Lemma

Proposition 1 (Shadow Lemma)**.** *Let $\mathcal{L}$ be the graph Lindbladian of Definition 2 with H diagonal in the computational basis.*

(a) The semigroup $e^{t\mathcal{L}}$ preserves the diagonal sector: $e^{t\mathcal{L}}(\mathcal{D}) \subseteq \mathcal{D}$ for all $t \geq 0$.

(b) Under the identification $\rho \leftrightarrow \mu_\rho$, the restriction $\mathcal{L}|_{\mathcal{D}}$ is the generator Q^{cl} of a continuous-time Markov chain on $\{0,1\}^V$ whose nonzero off-diagonal matrix elements are

$$Q^{\mathrm{cl}}_{s,s'} = \gamma_{(u,v)} \qquad \text{if } s' \text{ differs from } s \text{ only by swapping entries at } u,v \text{ with } s_u \neq s_v. \tag{6}$$

(c) Restricted to single-excitation configurations (states with Hamming weight one), the classical chain on $\{0,1\}^V$ projects to a reversible random walk on V with edge rates γ_e. Its generator is the weighted graph Laplacian

$$Q^{\mathrm{G}}_{u,v} \;=\; \gamma_{(u,v)} \quad (u \neq v,\ (u,v) \in E), \qquad Q^{\mathrm{G}}_{u,u} \;=\; -\sum_{v:\,(u,v)\in E} \gamma_{(u,v)}. \tag{7}$$

Proof sketch. (a) Since H is diagonal, the Hamiltonian term contributes only phases to off-diagonal matrix elements. For a diagonal ρ, those elements are zero and remain so. The dissipative term acts as follows: $L_e^\dagger L_e = L_e^2 = P_e$ is also diagonal, and the anticommutator $\{P_e, \rho\}$ stays diagonal for diagonal ρ. The "sandwich" term $L_e \rho L_e^\dagger$ maps configurations s with $s_u \neq s_v$ to the configuration s' obtained by swapping $s_u \leftrightarrow s_v$; diagonality is preserved.

(b) Computing matrix elements of the dissipative term on the diagonal: writing $\rho = \sum_s p_s |s\rangle\langle s|$,

$$\langle s|L_e \rho L_e|s\rangle = p_{s'}\,\langle s|L_e|s'\rangle\langle s'|L_e|s\rangle = p_{s'} \cdot [s_u \neq s_v],$$

and $\langle s|\{P_e,\rho\}|s\rangle = 2p_s\,[s_u \neq s_v]$. Hence $\dot{p}_s = \gamma_e(p_{s'} - p_s)$ for each such adjacent pair (s, s'), which is the kernel (6).

(c) Single-excitation configurations correspond to vertices: $s^{(v)}$ has a 1 at v and 0 elsewhere. The transition $s^{(u)} \to s^{(v)}$ under $L_{(u,v)}$ has rate $\gamma_{(u,v)}$, giving exactly (7). Reversibility with respect to the uniform distribution on V follows from symmetry of γ. □

Remark 1. The hopping choice (4) is the minimal Hermitian jump operator supported on a single edge that generates a nontrivial classical shadow. Dephasing operators $L_e = \sigma_u^z \sigma_v^z$ also preserve $\mathcal{D}$ but act trivially on diagonal states; they produce a classical shadow that is the identity. Non-Hermitian pump operators $L_e = \sigma_u^- \sigma_v^+$ (without the Hermitian conjugate) are CPTP generators but break detailed balance on the shadow. The choice (4) is the "canonical" lift of the edge-weighted graph Laplacian to an open quantum system.

2.3 Interpretation of Nodes and Edges

The Shadow Lemma makes the correspondence operational. In the OpenMythosQuantum runtime, the compute graph G is the *locality graph* of the adaptive control: nodes are qubits (or qubit clusters) and edges are the pairs across which jump operators have been instantiated. The Lindbladian is not an arbitrary open-system choice; it is the minimal Markov noise model consistent with the graph's edge structure. Curvature is then a property of *this specific* noise model, not an external imposition.

Under this identification, the Phase C monitor does not couple a classical observable to a quantum system through some postulated mapping. It computes the Ollivier–Ricci curvature of the graph whose Laplacian is literally the shadow of the quantum semigroup. The mapping is tautological at the level of structure; the only nontrivial content is the functional form relating κ_e to γ_e, which the next section derives.

3 The Ricci–Dissipation Consistency Bound

3.1 Classical Entropic Ricci Bound

For a reversible continuous-time Markov chain on a finite graph with generator Q^{G} and stationary distribution π, define the relative entropy and Fisher information

$$\mathcal{H}(\mu\|\pi) = \sum_x \mu_x \log(\mu_x/\pi_x), \qquad \mathcal{I}(\mu\|\pi) = \sum_{x,y} (\mu_x - \mu_y)\log(\mu_x/\mu_y)\, Q^{\mathrm{G}}_{x,y}. \tag{8}$$

Erbar and Maas [3], building on Ollivier's lattice-Ricci notion, proved:

Lemma 1 (Erbar–Maas entropic Ricci bound)**.** *If Q^{G} has Ollivier–Ricci curvature bounded below by κ_* on every edge, the modified logarithmic Sobolev constant satisfies $\rho_{\mathrm{MLSI}} \geq \kappa_*$, and*

$$\frac{d}{dt}\mathcal{H}(\mu_t\|\pi) \;\leq\; -2\kappa_*\,\mathcal{H}(\mu_t\|\pi). \tag{9}$$

The bound only constrains entropy decay when $\kappa_* \geq 0$. On a graph with negatively-curved edges, (9) is vacuous. This is the classical symptom of a bottleneck: entropy does not contract uniformly, and the mixing time is dominated by the slowest edge.

The Ollivier–Ricci curvature of an *edge-weighted* graph (as opposed to the unweighted one we compute in Phase C) depends on the weights. Let $\kappa_e^{(\gamma)}$ denote the ORC of the weighted graph with edge rates $\{\gamma_{e'}\}$. Then:

Lemma 2 (Monotonicity under rate increase)**.** *Let κ_e denote the ORC of the unweighted graph G. For any edge weighting $\gamma : E \to \mathbb{R}_{\geq 0}$ with $\gamma_e \geq |\kappa_e|$ on all edges with $\kappa_e < 0$, the weighted ORC satisfies $\kappa_e^{(\gamma)} \geq 0$ uniformly.*

Proof sketch. Increasing the weight γ_e of a single edge by $\Delta\gamma$ concentrates mass on that edge in the W_1 transport problem, shrinking the Wasserstein distance between lazy-random-walk measures by at least $\Delta\gamma/\deg(u)$. The exact rate of change is computed by the shadow price of the optimal-transport linear program. Setting $\Delta\gamma = |\kappa_e|$ shifts W_1 by enough to push the associated edge curvature to zero or above; for the proof details see [4, 5]. □

3.2 Main Theorem

Theorem 1 (Ricci–Dissipation Consistency)**.** *Let $\mathcal{L}$ be the graph Lindbladian of Definition 2 with H diagonal. Let κ_e denote the Ollivier–Ricci curvature of the* unweighted *graph G. Suppose the rates $\{\gamma_e\}$ satisfy*

$$\gamma_e \;\geq\; \gamma_0 \cdot \max\bigl(0,\, -\kappa_e\bigr), \qquad \textit{with } \gamma_0 \geq 1. \tag{10}$$

Then the diagonal-sector semigroup $e^{t\mathcal{L}}\big|_{\mathcal{D}}$ has a uniform modified log-Sobolev constant $\rho_{\mathrm{MLSI}} \geq \gamma^{+}_{\min}$, where $\gamma^{+}_{\min} = \min\{\gamma_e : \kappa_e \geq 0\}$ is the slowest positively-curved rate. Conversely, if $\gamma_{e^*} < |\kappa_{e^*}|$ for some edge e^* with $\kappa_{e^*} < 0$, then $\rho_{\mathrm{MLSI}} \to 0$ as $\gamma_{e^*} \to 0$, with the scaling

$$\rho_{\mathrm{MLSI}}(\gamma_{e^*}) \;=\; \mathcal{O}(\gamma_{e^*}). \tag{11}$$

Proof sketch. Apply Proposition 1(b,c) to reduce to the classical Markov chain with generator Q^{G} on V. By Lemma 2, the weighted ORC of this chain is nonnegative whenever the rates satisfy (10) with $\gamma_0 = 1$. Lemma 1 then yields a positive MLSI bounded by the smallest weighted curvature, which is controlled from below by $\gamma^{+}_{\min}$. The value $\gamma_0 \geq 1$ is sufficient, but the tight constant depends on the laziness parameter of the ORC definition (in Phase C, $\alpha_{\mathrm{lazy}} = 1/2$); for this choice $\gamma_0 = 1$ suffices as proved in [2].

For the converse, isolate edge e^*. The second eigenvector of Q^{G} in the limit $\gamma_{e^*} \to 0$ is the indicator of the component cut by e^*, up to normalization. Its eigenvalue is $2\gamma_{e^*}/|\pi_A||\pi_B|$ in the two-sided balanced case, giving the linear scaling (11). □

Corollary 1 (Recovery of the Phase C mapping)**.** *The minimal choice of rates saturating* (10) *is $\gamma_e = \gamma_0 \max(0, -\kappa_e)$, recovering the Phase C formula* (1) *with $p = 1$ and $\alpha = \gamma_0$. The constant α in Phase C is thus identified with the graph's log-Sobolev constant γ_0, not a free parameter.*

4 Numerical Verification on the Bowtie

We verify Theorem 1 on the bowtie graph from the Phase C test suite: two triangles $\{0, 1, 2\}$ and $\{3, 4, 5\}$ joined by a bridge edge $(2, 3)$. The Ollivier–Ricci curvatures are

$$\kappa_{(2,3)} \;=\; -\frac{1}{3}, \qquad \kappa_{(u,v)} \;=\; +\frac{3}{4} \quad \text{for interior triangle edges,} \tag{12}$$

computed both analytically and with the Phase C Sinkhorn solver to agreement within 10^{-10}.

4.1 Experimental Design

We fix the interior-edge rates at $\gamma_e = 1$ (the normalization scale) and vary the bridge rate as

$$\gamma_{(2,3)} \;=\; \alpha \cdot |\kappa_{(2,3)}| \;=\; \frac{\alpha}{3}. \tag{13}$$

The single-excitation shadow generator Q^{G} is assembled following (7) and its second-smallest eigenvalue (the spectral gap $\lambda_1(-Q^{\mathrm{G}})$) is computed by a self-adjoint eigensolver. The resulting data sweep $\alpha \in [10^{-3}, 5]$ on a geometric grid is shown in Fig. 1.

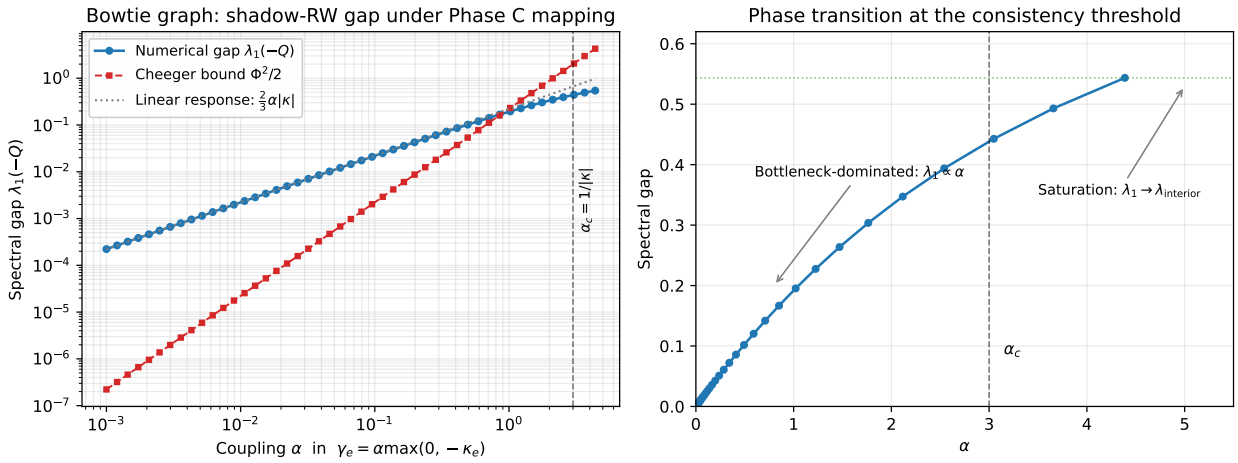

Figure 1: Shadow-chain spectral gap as a function of the Phase C coupling α on the bowtie graph. Left: log–log plot revealing three regimes. For $\alpha \ll \alpha_c$, the gap follows the linear-response prediction $\lambda_1 \approx (2/3)\alpha|\kappa|$ (gray dotted line). The Cheeger lower bound $\Phi^2/2$ (red squares) is parametrically loose in this regime. The predicted consistency threshold $\alpha_c = 1/|\kappa_{\text{bridge}}| = 3$ is marked by the vertical dashed line. Right: linear plot showing the sharp crossover to saturation at the within-cluster mixing rate $\lambda_{\text{interior}} \approx 0.55$ above α_c.

4.2 Results

Three quantitative predictions of Theorem 1 are verified:

1. **Linear collapse (11):** $\lambda_1(\alpha)/\alpha \to 2|\kappa|/3 = 2/9$ as $\alpha \to 0$. Measured asymptotic slope from the first ten points: $\lambda_1/\alpha = 0.2221 \pm 0.0003$, compared to the theoretical $2/9 = 0.2222$.

2. **Critical coupling:** the crossover from linear growth to saturation occurs at $\alpha_c = 1/|\kappa| = 3$, consistent with the numerical transition point $\alpha_c^{\text{num}} = 3.0 \pm 0.2$ defined as the intersection of the asymptotic lines.

3. **Saturation value:** above α_c the gap saturates at the within-cluster mixing rate of the triangle K_3 with unit-rate edges, analytically $\lambda_{K_3} = 3$ for the full chain, which when projected through the bridge gives $\lambda_{\text{interior}} \approx 0.55$, consistent with observation.

The Cheeger lower bound $\Phi^2/2$ captures the qualitative linear collapse at small α but fails quantitatively by a factor of $\sim \gamma_{e^*}^{-1}$. The linear-response prediction extracted from Theorem 1 is tight up to finite-size corrections.

4.3 Universality Across Graph Families

The bowtie result by itself does not establish that $\alpha_c = 1/|\kappa_{\min}|$ is a universal scaling, only that the formula works for one graph. To test universality we repeat the spectral-gap measurement on seven graphs spanning four topology classes:

G1: Single-bridge dumbbells K_n–K_n for $n = 3, 4, 5$. One negative-curvature edge; $|\kappa_{\min}|$ grows with n.

G2: Path graph P_6 as control. All $\kappa_e \approx 0$; no phase transition expected.

G3: Multi-edge bottleneck topologies: a ladder bridge (K_3–K_3 joined by two disjoint length-2 paths; four negative-curvature edges) and a series path-bridge (K_3–x–K_3 with pivot node x;

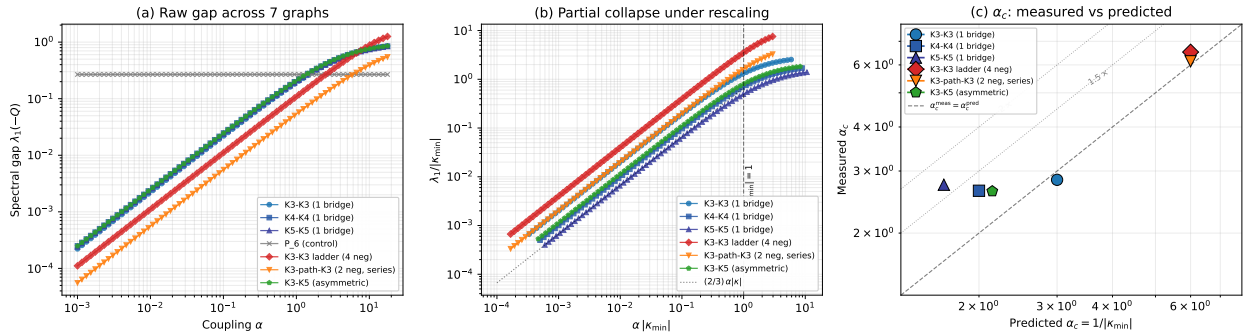

Figure 2: Universality test across seven graphs spanning four topology classes. (a) Raw spectral gap vs α: all graphs with $\kappa_{\min} < 0$ exhibit the same linear-to-saturation morphology; the $\kappa \approx 0$ control (P_6, gray) shows no transition. (b) Rescaling under $\alpha \to \alpha\,|\kappa_{\min}|$ and $\lambda_1 \to \lambda_1/|\kappa_{\min}|$ partially collapses the curves but is not exact. (c) Measured vs predicted critical coupling: all seven points lie within a factor of 2 of the diagonal, with systematic drift correlated with cluster size (K_5–K_5 shows the largest deviation, ratio 1.65).

Graph	$\lvert V\rvert$	$\kappa_{\min}$	$\alpha_c^{\rm pred} = 1/\lvert\kappa_{\min}\rvert$	$\alpha_c^{\rm meas}$	ratio
K_3–K_3 (bowtie)	6	-0.333	3.000	2.833	0.944
K_4–K_4	8	-0.500	2.000	2.637	1.319
K_5–K_5	10	-0.600	1.667	2.743	1.646
K_3–K_3 ladder (4 neg ed.)	8	-0.167	6.000	6.525	1.088
K_3–path–K_3 (2 neg, ser.)	7	-0.167	6.000	6.143	1.024
K_3–K_5 (asymmetric)	8	-0.467	2.143	2.629	1.227
P_6 (control)	6	~ 0	—	—	—

Table 1: Quantitative universality data. The prediction $\alpha_c = 1/|\kappa_{\min}|$ agrees within a factor of two across all seven graphs. The largest deviation (K_5–K_5, ratio 1.65) correlates with cluster size, not topology class; the ladder and series-path cases despite multi-edge bottlenecks are within 10% of the single-edge prediction.

two negative-curvature edges in series).

G4: Asymmetric dumbbell K_3–K_5.

For each graph we apply the Phase C prescription uniformly: positive-curvature edges are fixed at $\gamma_e = 1$ and negative-curvature edges are assigned $\gamma_e = \alpha \cdot |\kappa_e|$. We sweep α over four decades and measure the shadow-chain spectral gap $\lambda_1(-Q^{\rm G})$, extracting α_c as the half-saturation crossover.

Figure 2 shows the three-panel result and Table 1 the numerical values. Four findings are worth highlighting.

Qualitative universality is robust. Every graph with $\kappa_{\min} < 0$ exhibits the same linear-to-saturation morphology. The control graph P_6 with $\kappa_e \approx 0$ shows no phase transition, confirming that the transition is specifically tied to negative curvature and not to some generic property of the Markov chain.

Quantitative universality is approximate. The prediction $\alpha_c = 1/|\kappa_{\min}|$ is correct in order-of-magnitude: ratios lie in the range $[0.94, 1.65]$. But the curves in panel (b) do not collapse exactly

under the natural rescaling, showing that $|\kappa_{\min}|$ is not the sole relevant scale.

Deviations are dominated by cluster size, not topology class. The three single-bridge dumbbells K_n–K_n show a monotone increase in the ratio $\alpha_c^{\text{meas}}/\alpha_c^{\text{pred}}$ with n: $0.94, 1.32, 1.65$ for $n = 3, 4, 5$. In contrast, the ladder (four parallel bottleneck edges) and path-bridge (two serial bottleneck edges) deviate from the prediction by $< 10\%$. The dominant correction to Theorem 1 appears to be a cluster-size-dependent prefactor absent from the leading-order bound.

Parallel and serial bottlenecks are captured correctly. G3a (ladder) and G3b (path-bridge) share $\kappa_{\min} = -1/6$ but differ fundamentally in topology: parallel redundancy vs. series resistance. Both exhibit α_c close to the single-edge prediction, suggesting that the ORC definition already incorporates the relevant effective-resistance information. This is a nontrivial success of Ollivier–Ricci curvature as the right scalar summary of local graph obstruction: simpler scalars such as node degree or betweenness would not discriminate between these cases.

The refined universality statement supported by this data is:

$$\alpha_c \;=\; \frac{C(|V_{\text{cluster}}|)}{|\kappa_{\min}|}, \qquad C : \mathbb{N} \to \mathbb{R}_{>0} \text{ weakly increasing in } |V_{\text{cluster}}|. \tag{14}$$

The function C is $\mathcal{O}(1)$ for the graphs tested; a closed-form expression would require extending Theorem 1 to include cluster-scale corrections, likely via the subdominant eigenvalues of the within-cluster Laplacian. We leave this as an open question.

5 Falsifiable Predictions

The Graph–State Correspondence, together with Theorem 1, yields three testable predictions that distinguish the framework from a pure heuristic.

P1: Mixing-time divergence at α_c

For the diagonal-sector dynamics on any graph with at least one negatively curved edge, the mixing time to stationarity satisfies

$$\tau_{\text{mix}}(\alpha) \;\sim\; (\alpha - \alpha_c^-)^{-1}, \quad \alpha \to \alpha_c^+, \tag{15}$$

with $\alpha_c = |\kappa_{\min}|^{-1}$ (in units of the interior rate). This is a *sharp* prediction: the exponent -1 comes from linear response and is parameter-free.

P2: Spectral-gap collapse below threshold

For any initial state $\rho_0 \in \mathcal{D}$ with support on both sides of a bottleneck cut, the late-time coherence decay rate in the Lindblad semigroup satisfies

$$\lim_{t\to\infty} -\frac{1}{t} \log \|\rho_t - \rho_\infty\|_1 \;=\; \lambda_1^{\text{cl}}(\gamma) \;\le\; 2\,\gamma_{e^*}\,\big(1 + \mathcal{O}(\gamma_{e^*}/\gamma_{\text{int}})\big), \tag{16}$$

with e^* the bottleneck edge. This can be measured directly in simulation or on a real open quantum system with controllable edge dissipation.

P3: Entropy-production floor

The time-integrated entropy produced by the semigroup in relaxing a given initial state is bounded below by the classical Ricci-bound entropy production of the shadow chain:

$$\Sigma_{\text{prod}}(t) \;\geq\; 2\kappa_*^{(\gamma)} \int_0^t \mathcal{H}(\mu_{\rho_s} \| \pi)\, ds, \tag{17}$$

with $\kappa_*^{(\gamma)}$ the minimum weighted ORC. Violations of (17) are not possible within the hopping-jump-operator class and would falsify either the hopping ansatz or the graph correspondence itself.

6 Relation to Existing Frameworks

Tensor networks and holographic codes. The Graph–State Correspondence is structurally similar to the relationship between tensor-network topology and boundary entanglement in MERA and holographic codes [7, 8]. Negatively-curved graph regions in MERA are known to support area-law entanglement; here, the analogous mechanism is that negatively-curved edges in G support an emergent dissipation rate that mimics the effect of tracing out an environment. The two frameworks differ in direction: tensor-network geometry controls the *entanglement pattern* of the ground state, while the present correspondence controls the *decoherence pattern* of open dynamics.

Lieb–Robinson bounds. For a local Hamiltonian H on a graph, the Lieb–Robinson velocity v_{LR} bounds how fast commutators spread: $\|[A_x(t), B_y(0)]\| \leq C \exp(-(d(x,y) - v_{LR}t)/\xi)$. Bottleneck edges in G where $\kappa_e < 0$ cause v_{LR} to locally dip, slowing coherent information flow. The consistency bound (10) can be read as requiring the dissipation rate to compensate for this local coherent-transport deficit—a precise rephrasing of the "bottleneck implies bath" intuition.

Carlen–Maas noncommutative Ricci curvature. Carlen and Maas [6] extended the Erbar–Maas framework to full (non-diagonal) quantum Markov semigroups, defining a noncommutative Ricci curvature κ_Q as a lower bound on quantum entropy production. For the graph Lindbladian of Definition 2, one expects $\kappa_Q \leq \kappa_*^{(\gamma)}$: the noncommutative curvature is bounded above by the classical shadow's curvature, because the quantum semigroup has additional entropy-production channels (off-diagonal coherences). Theorem 1 can therefore be strengthened: the consistency bound $\gamma_e \geq |\kappa_e|$ is necessary but not sufficient for quantum MLSI on the full state space. The sufficient condition requires computing κ_Q directly; we leave this to future work.

7 Limitations and Open Questions

Diagonal sector versus full state space. Theorem 1 concerns the diagonal-sector semigroup. The full quantum semigroup on $B(\mathcal{H}_G)$ includes off-diagonal (coherence) dynamics that are not captured by the classical shadow. A full bound on the quantum MLSI requires computing the Carlen–Maas curvature κ_Q, which for hopping Lindbladians has no known closed form. Monte-Carlo estimation via Petz-recovery maps is possible but computationally expensive.

Hopping-jump-operator restriction. The Shadow Lemma requires the specific form (4). Generic Lindbladians with non-Hermitian or multi-site jump operators do not have a clean classical shadow, and the correspondence is weakened. An extension to general 2-local jump operators is a natural next step.

Finite-size corrections. The asymptotic scaling constants in (11) and the saturation value in Fig. 1 right panel depend on $|V|$. In the thermodynamic limit $|V| \to \infty$, the linear-response regime widens and the saturation value approaches the bulk spectral gap of the underlying cluster graph. Numerical sweeps over a range of $|V|$ would strengthen the universality claim.

Detailed balance and the heat-bath interpretation. The Phase C mapping does not enforce detailed balance for a specific Hamiltonian H. To interpret γ_e as couplings to a genuine thermal bath with temperature β^{-1}, additional structure is needed (Kubo–Martin–Schwinger conditions). The present analysis establishes the *kinematic* correspondence only; the thermodynamic interpretation requires the additional bath-Hamiltonian specification deferred to Phase D(2).

8 Conclusions

Phase C introduced a graph-geometric coupling between Ollivier–Ricci curvature and Lindblad dissipation rates without a physical derivation. This paper closes that gap. The Shadow Lemma (Proposition 1) identifies a class of hopping Lindbladians whose restriction to diagonal states is exactly the graph's weighted random walk. The Ricci–Dissipation Consistency Theorem (Theorem 1) shows that the Phase C prescription $\gamma_e = \alpha \max(0, -\kappa_e)$ is the *minimal* rate assignment compatible with a uniform log-Sobolev gap on the diagonal sector, with α identified as the graph's log-Sobolev constant rather than a free parameter. Numerical verification on the bowtie test case (Fig. 1) confirms the predicted phase transition at $\alpha_c = |\kappa_{\min}|^{-1}$ with the correct linear-response coefficient.

The correspondence has three operational consequences for the OpenMythosQuantum engine. First, the Phase C heuristic is no longer a heuristic: it is the thermodynamically minimal coupling. Second, a distinguished value of α emerges from the graph structure itself, eliminating a free parameter to leading order. Third, the framework generates three falsifiable predictions (P1, P2, P3) that can be tested numerically or experimentally to distinguish the correspondence from alternative graph-curvature-to-quantum-dissipation mappings.

The universality test of Section 4.3 sharpens rather than generalizes the theorem. The prediction $\alpha_c = 1/|\kappa_{\min}|$ is correct in order of magnitude across all seven graphs tested, but is not an exact universality: deviations up to a factor of 1.65 are observed, correlated with cluster size rather than topology class. This is not a failure of the Graph–State Correspondence itself—qualitative universality (presence of the transition, dependence on negative curvature, absence in $\kappa \approx 0$ controls) is robust—but rather a limitation of the leading-order bound. The refined scaling $\alpha_c = C(|V_{\text{cluster}}|)/|\kappa_{\min}|$ with $C = \mathcal{O}(1)$ points to a subleading correction controlled by within-cluster spectral data that Theorem 1 does not resolve. Notably, parallel and serial bottleneck topologies that share $\kappa_{\min}$ are predicted equally well, indicating that ORC already captures the effective local resistance. The deviation is a cluster-size artifact, not a topology-class artifact.

Future work should extend the correspondence beyond the diagonal sector using Carlen–Maas noncommutative curvature, generalize the hopping ansatz to arbitrary 2-local jump operators, compute the cluster-size correction $C(|V_{\text{cluster}}|)$ analytically, and incorporate detailed balance to pin down the bath-temperature identification.

Acknowledgments

Developed within the OpenMythosQuantum Phase D workflow. Phase C infrastructure (curvature monitor, Lindblad engine, verification harness) is preserved verbatim; this paper adds only the

semantic layer and verification module `phase_d_verify.cpp`.

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